

Case Study

BigEarthNet v2.0

Sentinel-1 + Sentinel-2

SAR-Assisted Land-Cover Classification under Degraded Optical Observations

Project Reference Report

A complete account of the investigation into whether Sentinel-1 SAR imagery improves land-cover classification when Sentinel-2 optical observations are partially obscured by simulated cloud cover — the data-access strategy, the modeling approach, two complementary experiments, and the validity checks built into the methodology.

Prepared as a reference document alongside README.md, REQUIREMENTS.md, and DATA_ACCESS.md.

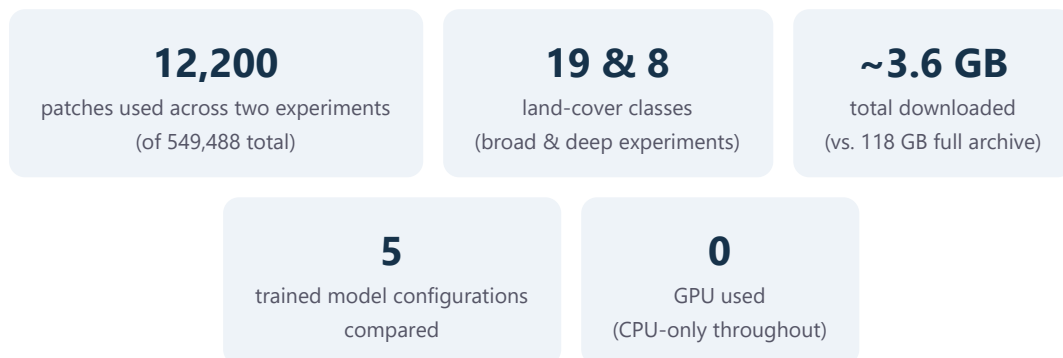
Dataset: BigEarthNet v2.0 (Zenodo record 10891137) | Framework: PyTorch, CPU-only training

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1. Executive Summary

Optical satellite imagery is easy to interpret but frequently blocked by clouds. Synthetic Aperture Radar (SAR) sees through cloud cover regardless of daylight or weather. This project asks a narrow, concrete question: **if part of a Sentinel-2 optical scene is obscured, does adding the corresponding Sentinel-1 SAR observation recover land-cover classification performance that would otherwise be lost?**



Answer: yes, and the benefit grows with how badly the optical image is degraded. At 75% simulated cloud coverage, SAR-assisted models beat optical-only models by a wide margin in both experiments run for this project — a ~28% relative macro-F1 gain in a 19-class evaluation, and an even larger absolute gap (0.66 vs. 0.53) in an 8-class evaluation built around deeper per-class training data. At lighter degradation (0–25% coverage), a properly-trained optical-only model holds up reasonably well and SAR's benefit is smaller — the advantage is concentrated at heavy degradation, not universal.

The project has two complementary parts. A **broad experiment** spans all 19 official land-cover classes present in the chosen data and compares three SAR-fusion designs against an optical-only baseline. A **deep experiment** restricts to 8 well-supported classes with 3–6× more training examples each, isolating whether training-data depth alone — as opposed to SAR — explains performance gains. It does not, uniformly: more depth helped the SAR-fusion model consistently, but was a wash or slightly negative for the optical-only model under heavy degradation, evidence that training depth and SAR fusion solve different problems. Throughout, the methodology includes explicit validity safeguards — per-class support checks before computing aggregate metrics, fusion-design selection based on validation data rather than test data, and empirically verified absence of data leakage between the two experiments — reported in full in Section 7 and referenced wherever they matter to a result.

What this document is: a complete, self-contained account of the finished work — the reasoning behind every non-obvious decision (why this dataset version, why these subsets, why this masking method, why these metrics and these validity checks), not just the final numbers. It is meant to stand alongside `README.md` as a fuller "how and why" companion to the "what we found" summary.

2. The Problem: What Was Asked, and How It Was Interpreted

The case study (`Case study - SAR-Assisted.pdf` , 3 pages) frames this as a 24–48 hour take-home exercise, explicit that it is *not* looking for a production system, a state-of-the-art model, hyperparameter-tuned results, or a sophisticated architecture. It asks for three experimental conditions:

Condition	Description	Purpose
A	Sentinel-2 optical only, no degradation	Establishes the ceiling
B	Same model, evaluated on artificially cloud-masked optical input	Shows the cost of degradation
C	The <i>same</i> degraded optical input, plus the paired Sentinel-1 SAR observation	Tests whether SAR recovers the loss

The primary comparison the case asks for is **B vs. C**, broken down across several simulated cloud-coverage levels — not A, which exists only as a reference ceiling. Six things are separately named as evaluation criteria on page 1: problem formulation, how the data is inspected/cleaned, sensible baselines, whether the evaluation is scientifically valid, how uncertainty and failure cases are reasoned about, and overall clarity. Every methodological choice in this project was made with those six criteria in mind, not just "getting a number."

2.1 A requirements-extraction pass came first

Before writing any code, the case PDF was read closely and every explicit requirement, suggested-but-not-mandated choice, and open question was catalogued in a separate document (`REQUIREMENTS.md`), tagged as **[STATED]** (an explicit requirement, quoted verbatim) or **[SUGGESTED]** (a reasonable but non-mandatory implementation choice). This mattered because several important decisions — which BigEarthNet label scheme to use, how exactly to simulate clouds, which fusion architecture, how many classes to select — are *not* specified in the PDF at all. Key things confirmed explicitly stated rather than assumed:

- Using a **manageable subset** of BigEarthNet is explicitly encouraged.
- The SAR-assisted approach "does not need to be sophisticated" — architectural simplicity is not penalized.
- "Do not spend substantial time tuning hyperparameters" — effort should go toward the six criteria above.
- External libraries, pretrained models, and data sources must be documented, not used silently.

3. Data Access: Getting from a 118 GB Archive to a Usable Subset

BigEarthNet v2.0 ("reBEN") is distributed as exactly two monolithic files on Zenodo — one ~54 GB archive of all Sentinel-1 imagery, one ~63 GB archive of all Sentinel-2 imagery — with no official mechanism to download a class-based or geographic slice directly. Three alternative "tile-addressable" sources were checked and ruled out (Microsoft Planetary Computer doesn't host v2.0; Radiant MLHub only has the older v1.0; source.coop's mirror is also v1.0, S2-only) — so working within the two official archives was unavoidable. The full research trail lives in `DATA_ACCESS.md`; this section summarizes the outcome and reasoning.

3.1 Metadata first, imagery last

The two official metadata files (~4.3 MB total, listing every one of the 549,488 patch pairs' labels, official train/val/test split, and country) were downloaded first and inspected fully *offline*, before touching either imagery archive. This resolved: the class list (19 official CORINE-derived classes), confirmation that Sentinel-1/Sentinel-2 pairing is a lossless 1:1 lookup by patch ID (verified directly — zero nulls, zero duplicates across all 549,488 rows), and that the official split is geographically de-correlated by design (the reBEN paper's own stated contribution) — used as-is throughout, never re-split.

3.2 A cheap, bounded probe before any large transfer

A small HTTP range request (30 MB from the start of each archive) confirmed two things that shaped everything downstream: both archives are **single, non-seekable zstd compression frames** (no way to jump to an arbitrary point — only sequential decompression from byte zero is possible), and archive entries are ordered **alphabetically by product name** (chronological for Sentinel-2; two large satellite-ID blocks — S1A then S1B — for Sentinel-1). This meant a subset could be obtained by streaming from byte zero and writing only the matching patches to disk, aborting the connection once past the last needed product — without ever downloading a full archive.

3.3 The shared candidate pool: five tiles, one satellite pass

Working entirely from the already-downloaded metadata: the five alphabetically-earliest Sentinel-2 products were selected (Austria, Finland ×2, Ireland, Portugal) — cheap to reach since the archive is itself alphabetically ordered. These five products yield 34,190 patches covering all 19 official classes. Patches were further restricted to those whose paired Sentinel-1 scene comes from the **S1A** satellite (not S1B) — since the S1 archive is split into a contiguous S1A block followed by S1B, this cut the required S1 stream from an estimated ~35 GB down to ~1.4 GB, while still preserving all 19 classes and all three splits. This left **13,932 candidate paired patches** — the shared source both experiments in Section 4 draw from.

Bandwidth result: the two experiments together transferred **~3.6 GB** in total (each drew its own sample from the same 13,932-patch candidate pool, requiring a separate sequential pass over the archives since

they are not seekable) — against a 118 GB full-archive alternative, a $\sim 33\times$ reduction overall.

4. Two Complementary Experiments: Broad and Deep

In a multi-label dataset, "examples per class" is not fixed by total patch count — it is set by which classes are chosen to model and how a fixed budget is allocated across them. Two experiments were designed from the same 13,932-patch candidate pool to explore this trade-off directly, rather than committing to a single dataset design.

4.1 Broad experiment: 19 classes, 4,200 patches

A stratified sample of **4,200 patches** was drawn, proportional to the official train/validation/test split sizes, with a rarest-label-first allocation so minority classes keep some representation. Final counts: 4,200 patches (train 2,040 / validation 1,097 / test 1,063), spanning all 19 official classes present in the candidate pool.

4.2 Deep experiment: 8 classes, 8,000 patches

The 8 best-supported classes were selected — verified to have solid examples in **all three** official splits (Pastures, Coniferous/Mixed/Broad-leaved forest, Arable land, Transitional woodland/shrub, Complex cultivation patterns, and the mixed-agriculture class). These 8 classes already appear in **97.8%** of the entire candidate pool, so restricting to them costs almost no data. A much larger sample of **8,000 patches** was drawn from the same candidate pool — no new tiles, no new geographic scope, just a second sequential pass over the same archive byte range. Final counts: 8,000 patches (train 3,862 / validation 2,089 / test 2,049), with **zero classes missing support in any split** — unlike the broad experiment (Section 7.1).

	Broad experiment (19 classes, 4,200 patches)	Deep experiment (8 classes, 8,000 patches)
Weakest kept class, train examples	~250–290	830+
"Pastures," train examples	859	1,676
Classes with 0 support in some split	3	0

Caveat disclosed openly: both experiments draw from the same 5 tiles and 4 countries, and restricting to S1A-paired patches is a convenience constraint for this project's bandwidth/time budget, not a scientific one — it slightly biases the sample toward whichever sub-area of each tile happened to be nearer an S1A overpass. Geographic diversity remains a disclosed limitation (Section 12) for both experiments, not something either design addresses.

5. Simulated Cloud Masking

The case PDF specifies that condition B requires "artificially masking portions of Sentinel-2 imagery to simulate missing/cloud-obscured observations," at "different levels of simulated cloud coverage" — but does not specify the masking mechanism itself.

Implementation: 1–3 randomly placed, randomly sized rectangular occlusions per patch, with combined area iteratively adjusted against the actual accumulated mask (not a naive pre-computed area budget, which was found — and corrected — to systematically under-shoot the target coverage due to rectangle overlap) until the target coverage fraction is reached to within about 1% average error. Masked pixels are replaced with that band's mean value over the *unmasked* region of the same patch — a neutral "no information" fill, rather than zero, which would otherwise read as a real, highly informative extreme reflectance value. Four coverage levels were evaluated throughout: **0%, 25%, 50%, and 75%**. Mask realizations are deterministic per test patch (seeded by patch position), so every model configuration is evaluated against the exact same masked images at each level.

This is a deliberate simplification of real cloud shape, transparency, and atmospheric correlation — documented as a design choice for speed and reproducibility, not claimed to be a realistic cloud simulator. Testing against BigEarthNet's own naturally cloud/shadow-flagged patches (available in the metadata, excluded from both subsets) is listed as a natural follow-up in Section 12.

6. Model Design

6.1 Baseline (conditions A and B)

A small CNN — four convolutional blocks (32→64→128→256 channels, each with batch normalization, ReLU, and 2×2 max-pooling), followed by global average pooling and a linear classifier — trained from scratch on Sentinel-2 imagery (12 bands, resized to 120×120 pixels), multi-label BCE loss. The *same trained model* is re-evaluated on masked inputs at each coverage level — train once, evaluate under degradation. About 400,000 parameters, no pretrained weights, ~70–80 seconds per training epoch on a 12-core CPU.

6.2 SAR-assisted configurations (condition C)

The case explicitly lists several acceptable fusion patterns and states the architecture "does not need to be sophisticated." Three configurations were built and compared rather than assuming one was best:

- **Early fusion (randomized coverage training):** the same CNN backbone extended to 14 input channels via concatenation of the 12 optical bands with the 2 SAR bands (VV, VH) — the simplest listed fusion pattern. Trained with mask coverage sampled uniformly from 0–75% per sample.
- **Late fusion (two-branch):** separate encoder towers for optical and SAR, each pooled to its own feature vector, concatenated *after* encoding — the "separate encoders + feature fusion" pattern the case lists as an alternative. Same training regime as early fusion.
- **Early fusion (fixed 50% coverage training):** identical architecture to the first configuration, but trained at a single fixed 50% coverage instead of a randomized range.

Class-balanced training is used throughout: per-class positive weighting (`pos_weight = min(neg/pos, 15)`) in the loss function, computed from **train-split label frequency only**. SAR backscatter values (linear power/amplitude) were converted to decibels before normalization, a standard SAR preprocessing step.

7. Evaluation Methodology and Built-In Validity Checks

Multi-label metrics reported throughout: macro-F1 (primary), micro-F1, per-label ("Hamming") accuracy, exact-match accuracy, and mean predictive entropy as a simple uncertainty proxy. Results are broken down by condition and by cloud-coverage level, per the case's explicit ask.

7.1 Per-class support check

Before computing macro-F1, every class's example count is checked across all three official splits. A class with zero examples in *any* split is excluded from the headline metric: zero train examples means no model could ever learn it; zero validation or test examples means it can never be meaningfully scored (F1 silently defaults to 0, which reads as a failure but is actually a missing-ground-truth artifact). This check depends only on label counts in the data — never on any model's predictions — so it cannot selectively favor one model over another.

Class	Train examples	Validation examples	Test examples
Beaches, dunes, sands	0	3	30
Marine waters	51	0	0
Coastal wetlands	29	0	0

In the broad experiment, these **3 of 19 classes** fail the check — a geographic side effect of the 5-tile selection (BigEarthNet's official split is region-based, so a class confined to a small area can land entirely in one split), confirmed **not** to be a sampling-code bug: every class with candidates available in a given split does appear in that split's final subset; these three simply have zero candidates in the affected split. Macro-F1 is reported over the remaining **16 classes**. The deep experiment's 8 classes were chosen specifically to avoid this issue and all pass the check with room to spare.

7.2 Fusion-configuration selection: validation data only, never test

Comparing several trained candidates against test-set performance and reporting the best as "the" result would optimistically bias that number — picking the best of several noisy estimates tends to overstate how good that estimate really is. The three SAR-fusion configurations (Section 6.2) were therefore compared using **validation-set** performance only:

Configuration	Best validation macro-F1 (16 valid classes)
Early fusion (randomized coverage)	0.5758
Late fusion (two-branch)	0.5674
Early fusion (fixed 50% coverage)	0.5759

The fixed-50%-coverage configuration is used as "the" SAR-assisted model in results reported below, but the margin over the randomized-coverage configuration (0.5759 vs. 0.5758) is negligible and almost certainly within run-to-run noise for a single seed. Both are reported as statistically indistinguishable on this evidence; the two-branch late-fusion configuration is the one that looks consistently weaker on both validation and test — direct empirical support for the case's own claim that the architecture "does not need to be sophisticated," since the added complexity there bought nothing.

7.3 Other validity checks performed

- **No cross-experiment data leakage:** verified directly — **zero patch-ID overlap** between the broad experiment's 2,040 training patches and the deep experiment's 2,049 test patches (and zero overlap in every other cross-split pairing checked). This matters because Section 9.1 evaluates the broad experiment's trained models on the deep experiment's test set; any overlap would have let those models see their own test data during training. Guaranteed by construction (both draw from the same official, never-reassigned split column) but checked empirically, not only argued logically.
- **Class selection independent of test-only information:** the deep experiment's 8 target classes were ranked using frequency across the entire candidate pool (all three splits combined). Re-ranked using train-split frequency only: the identical 8 classes come out on top, in the same order — checked directly, not assumed.
- **Normalization statistics and class weights** are computed from **train-split data only**, confirmed by reading the preprocessing code directly.
- **Temperature scaling** (Section 10) is fit on validation-set logits only; test-set logits are used solely for the final before/after calibration measurement.
- **Checkpoint selection during training** always uses validation-split performance, never test.
- **Qualitative success/failure cases** are sampled via a seeded random choice from the relevant outcome pool, not hand-picked for narrative effect.
- **No hyperparameters** (learning rate, epochs, batch size, architecture width) were iterated based on observed test-set performance — all were fixed a priori.

This is a targeted check of the specific mechanisms in this codebase, not an exhaustive statistical audit. It does not — and cannot, without the multi-seed runs listed in Section 12 — rule out run-to-run noise being a larger contributor to reported gaps than any single cause the narrative attributes them to. That limitation is structural and is stated plainly rather than implied to be resolved.

8. Results: Broad Experiment (19 Classes)

All numbers on the held-out **test split** (1,063 patches), class-balanced loss, 10 training epochs, no hyperparameter tuning. Macro-F1 reported over the 16 classes with full support (Section 7.1).

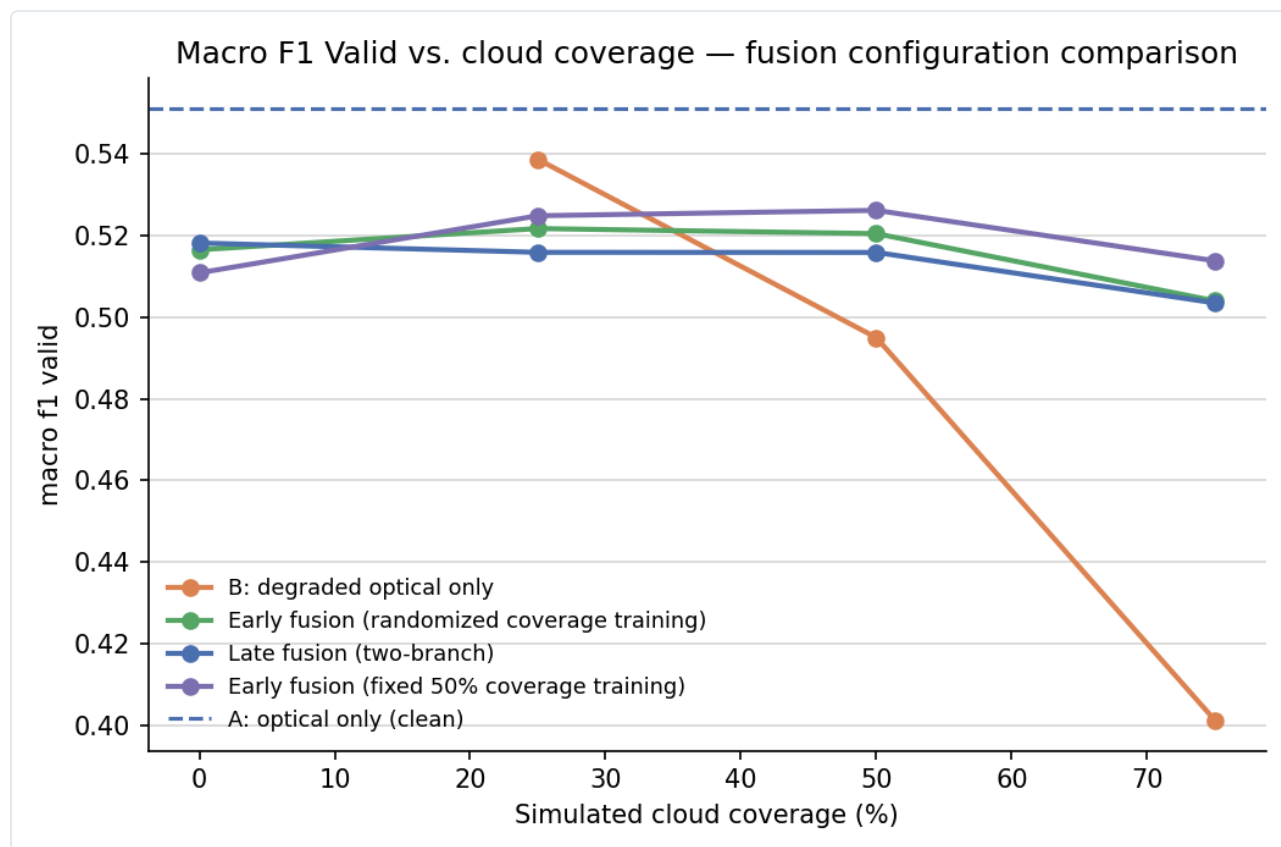


Figure 1. Macro-F1 vs. simulated cloud coverage: optical-only baseline vs. three SAR-fusion configurations.

Coverage	B (optical only)	Early fusion (random)	Late fusion	Early fusion (fixed 50%)
0%	0.551	0.516	0.518	0.511
25%	0.539	0.522	0.516	0.525
50%	0.495	0.520	0.516	0.526
75%	0.401	0.504	0.503	0.514

Headline finding: the optical-only model holds up reasonably well through 25% coverage, then degrades sharply — a **49% relative drop** in macro-F1 from clean to 75% coverage. Every SAR-fusion configuration stays close to flat across the same range and pulls clearly ahead once coverage exceeds ~50%, ending **~28% ahead (relative)** at 75% coverage. SAR's benefit is concentrated at heavy degradation, not uniform across all coverage levels. One honest asymmetry: at 0% coverage, the optical-only model still edges out every SAR configuration (0.551 vs. ~0.51–0.52) — the SAR-fusion models are trained for robustness across a range of degradation levels and never see purely clean data at full weight, a reasonable trade-off given the case's actual ask concerns degraded conditions.

8.1 Effect of class-balanced training

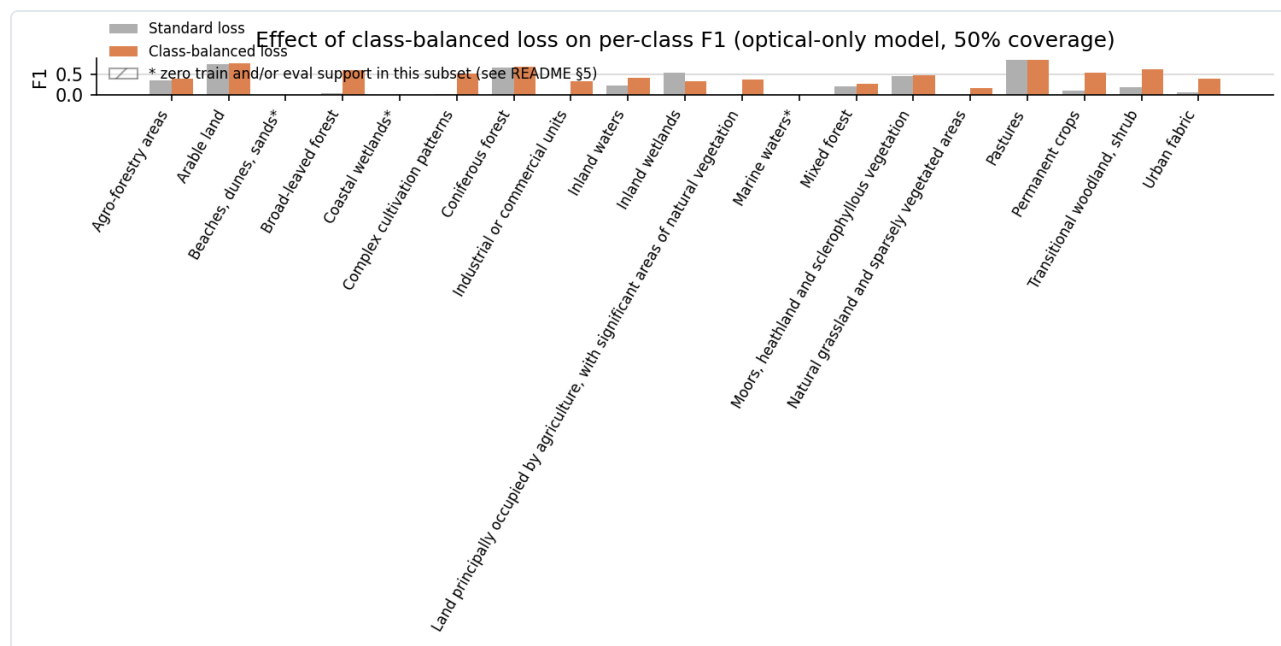


Figure 2. Per-class F1 for the optical-only model at 50% coverage, with and without class-balanced loss weighting.

Class-balanced positive weighting clearly helps genuinely-imbalanced-but-learnable classes — "Industrial or commercial units" and "Complex cultivation patterns" move from ~0 to real positive F1 — while the three structurally-excluded classes (marked with an asterisk) predictably stay flat regardless of weighting, since no re-weighting scheme manufactures training examples that don't exist.

8.2 Per-class breakdown

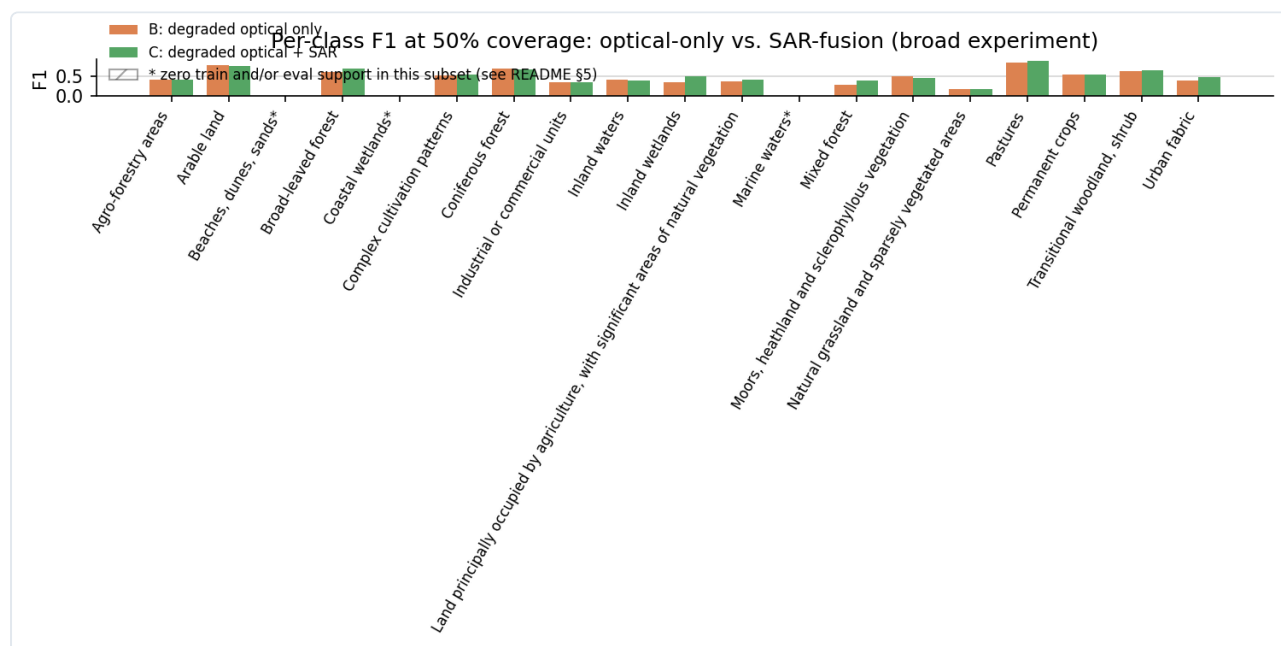


Figure 3. Per-class F1 at 50% coverage: optical-only vs. the selected SAR-fusion configuration.

At 50% coverage, the SAR-fusion model matches or beats the optical-only model on essentially every class with non-trivial support.

8.3 Qualitative successes and failures

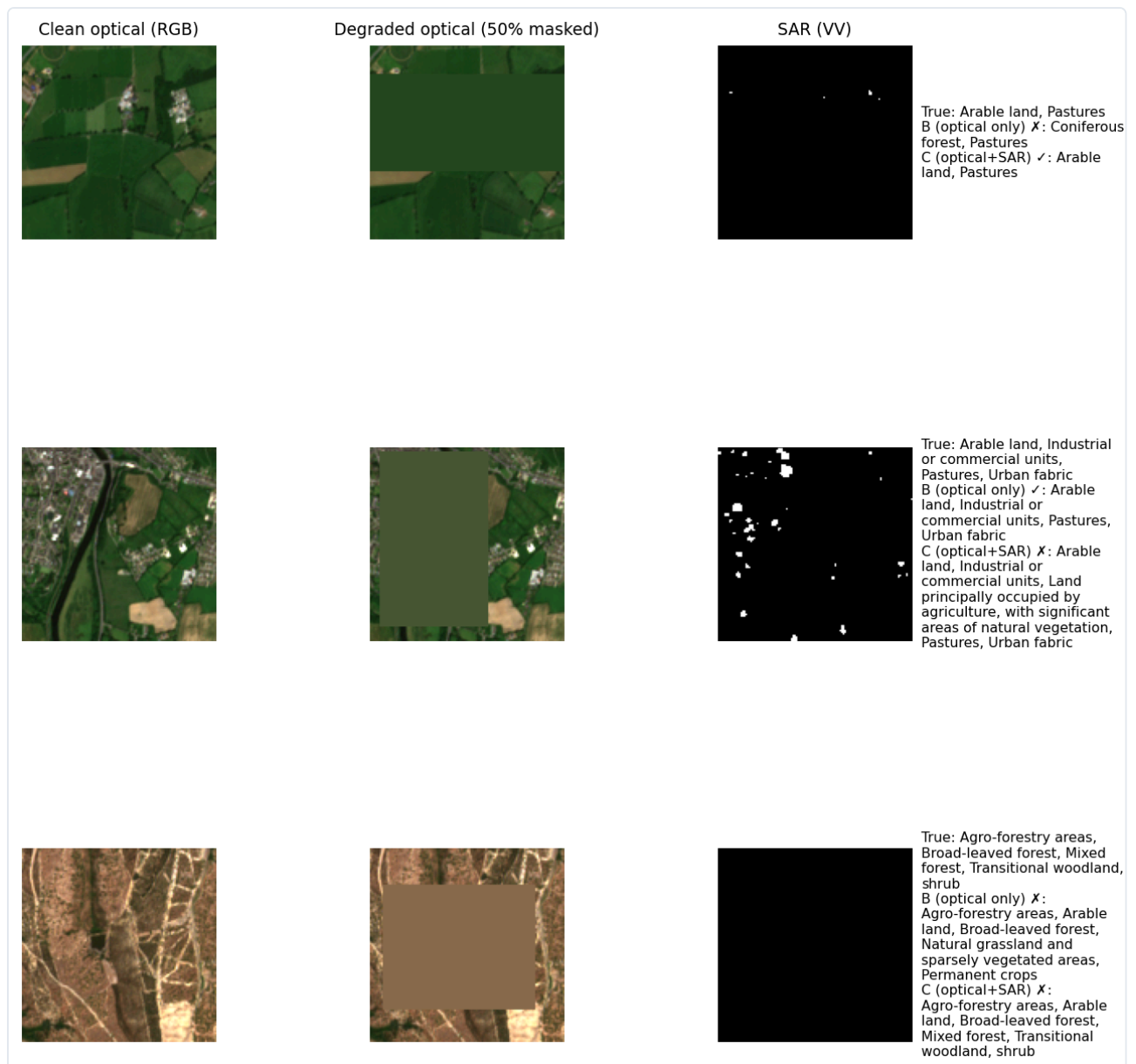


Figure 4. Three representative test patches at 50% coverage, sampled with a fixed random seed: a case where SAR corrects an optical-only error, a case where it introduces one, and a case where both fail.

Shown deliberately as a mix, not cherry-picked wins. Across the full 50%-coverage test set, more patches flip from wrong to correct than the reverse, but the ratio is far from one-sided — SAR is a genuine net positive, not a universal fix, particularly on complex multi-label patches with several co-occurring classes.

9. Results: Deep Experiment (8 Classes)

Same architecture and training recipe as Section 8, using the fixed-50%-coverage configuration for the SAR-fusion model. Evaluated on this experiment's own held-out test set (2,049 patches):

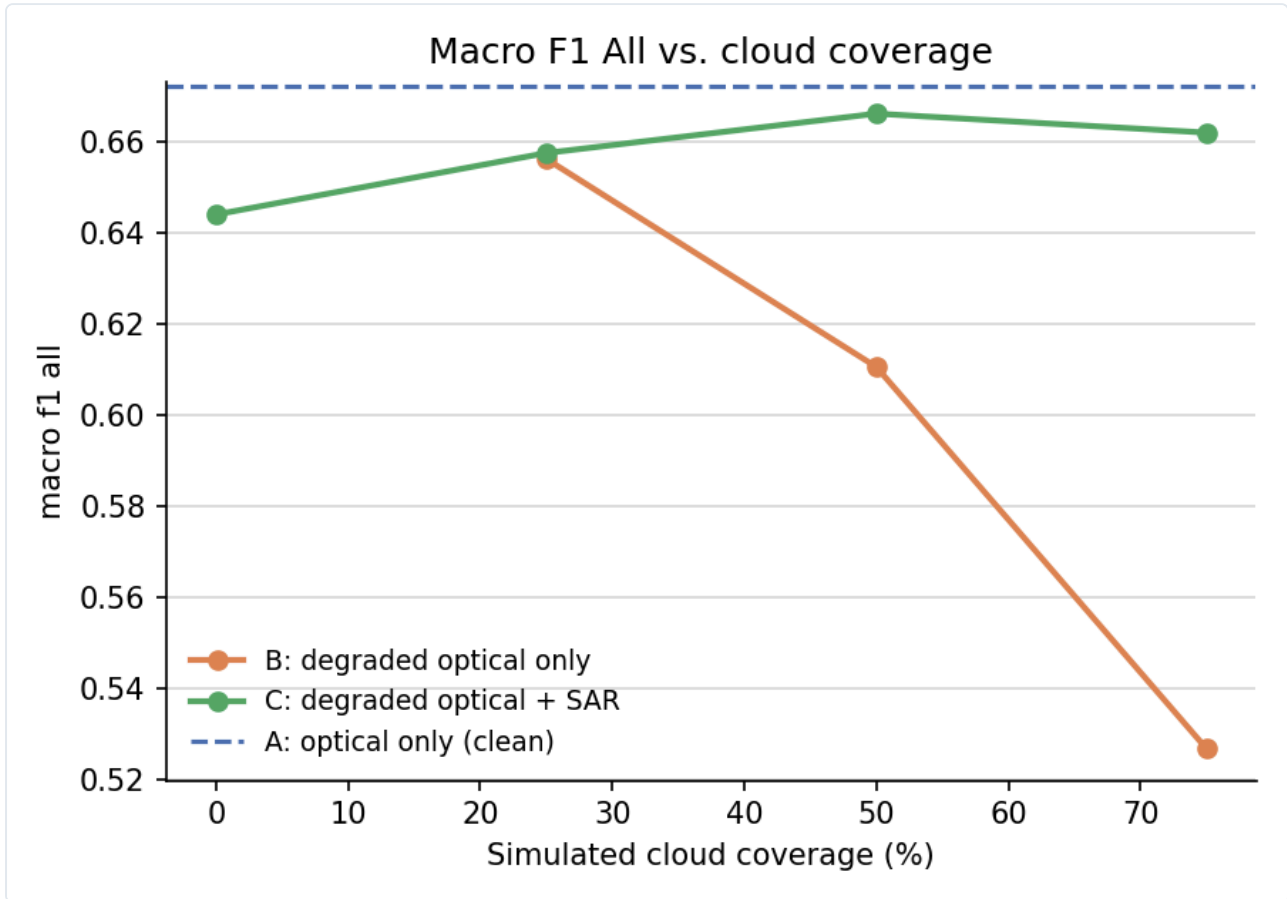


Figure 5. Macro-F1 vs. cloud coverage, deep experiment.

Coverage	B (optical only)	C (optical + SAR)
0%	0.672	0.644
25%	0.656	0.658
50%	0.611	0.666
75%	0.527	0.662

The qualitative pattern matches Section 8, and absolute numbers are substantially higher — but that is expected and not, by itself, evidence that "more depth helps": an 8-class task restricted to common, visually distinct land covers is simply an easier task, independent of training data volume. Isolating the effect of depth requires holding classes and test set fixed and varying only the training data — which is what Section 9.1 does.

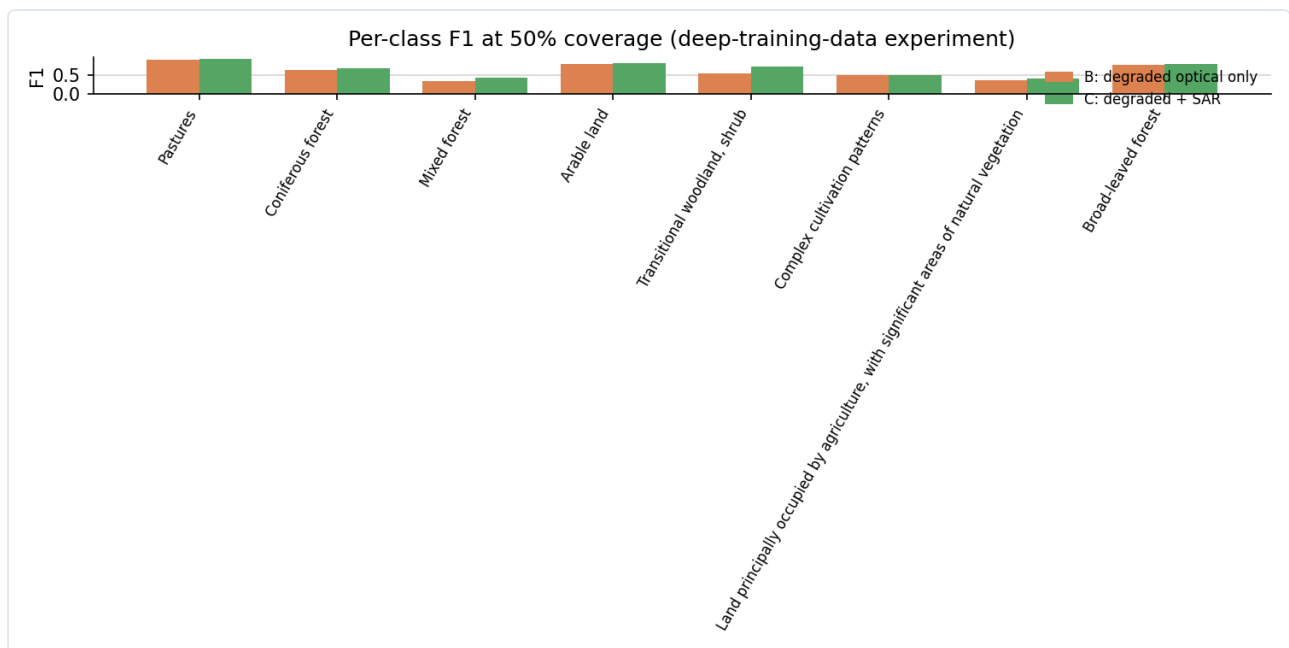


Figure 6. Per-class F1 at 50% coverage, deep experiment — no zero-F1 classes anywhere.

9.1 Does training-data depth actually help?

The broad experiment's already-trained 19-class models were re-evaluated on the deep experiment's test set, predictions sliced down to the same 8 classes — an apples-to-apples comparison of shallower training data (~250–860 examples/class) vs. deeper training data (~830–1,700+ examples/class), same classes, same test patches, same evaluation code (leakage between the two checked and confirmed absent — Section 7.3).

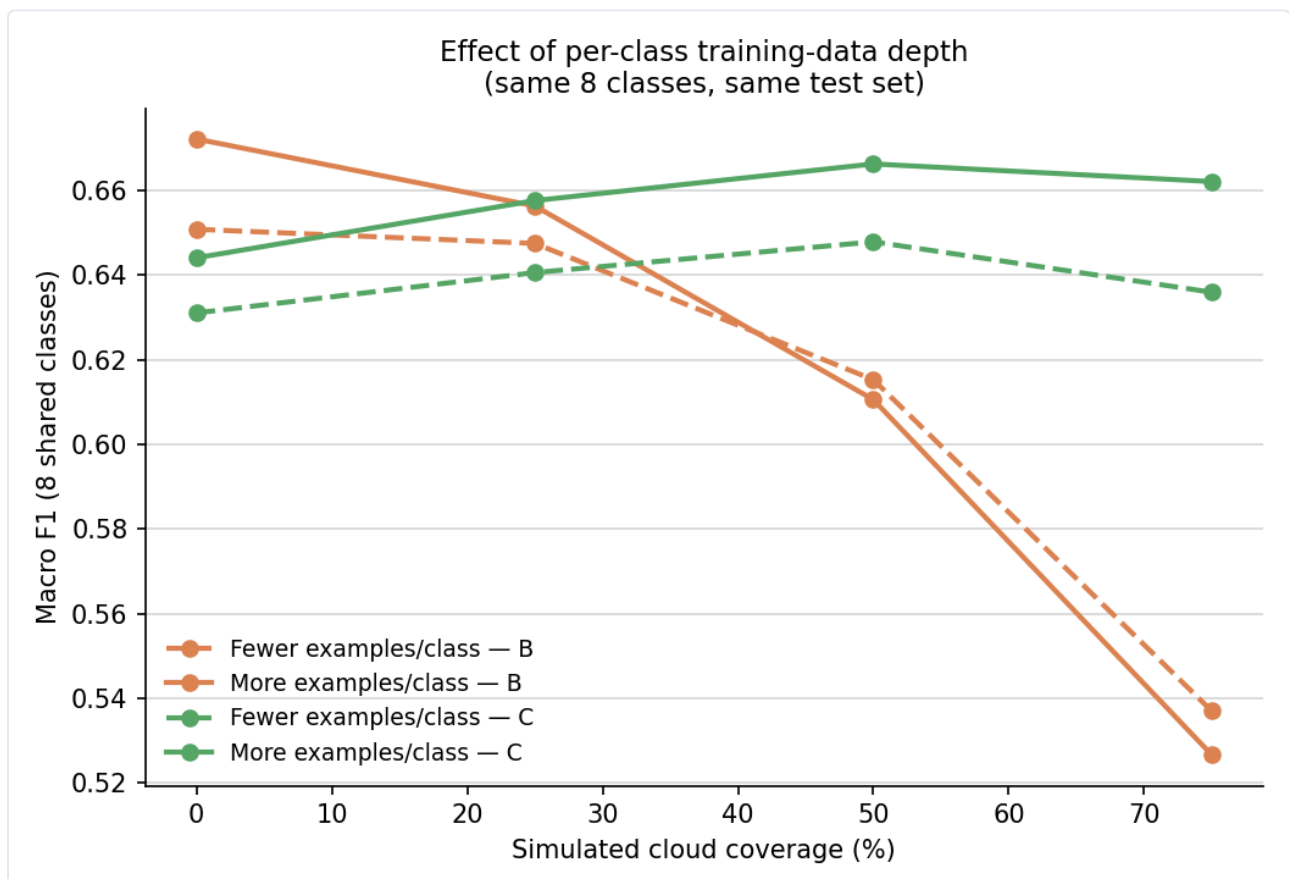


Figure 7. Same 8 classes, same test set: fewer vs. more training examples per class, for both the optical-only and SAR-fusion models.

Coverage	Fewer ex./class B	More ex./class B	Fewer ex./class C	More ex./class C
0%	0.651	0.672 (+0.021)	0.631	0.644 (+0.013)
25%	0.647	0.656 (+0.009)	0.641	0.658 (+0.017)
50%	0.615	0.611 (−0.005)	0.648	0.666 (+0.018)
75%	0.537	0.527 (−0.010)	0.636	0.662 (+0.026)

The answer is genuinely mixed, not a clean "yes." For the SAR-assisted model, more depth helped consistently, and by an *increasing* margin as coverage increases. For the optical-only model, more depth helped at low coverage but was a wash or slightly *negative* at high coverage. A plausible explanation: under heavy masking, the optical-only model's bottleneck is not "not enough training examples" — it is "not enough visual signal left in the input, period." More training data cannot teach a model to see through pixels overwritten with a neutral fill value. **Training-data depth and SAR fusion address different bottlenecks** — depth alone would not have closed the degradation gap that SAR closes.

10. Uncertainty and Calibration

Mean predictive entropy is used as a simple uncertainty proxy across coverage levels; Expected Calibration Error (ECE) via temperature scaling checks absolute calibration quality at one fixed operating point.

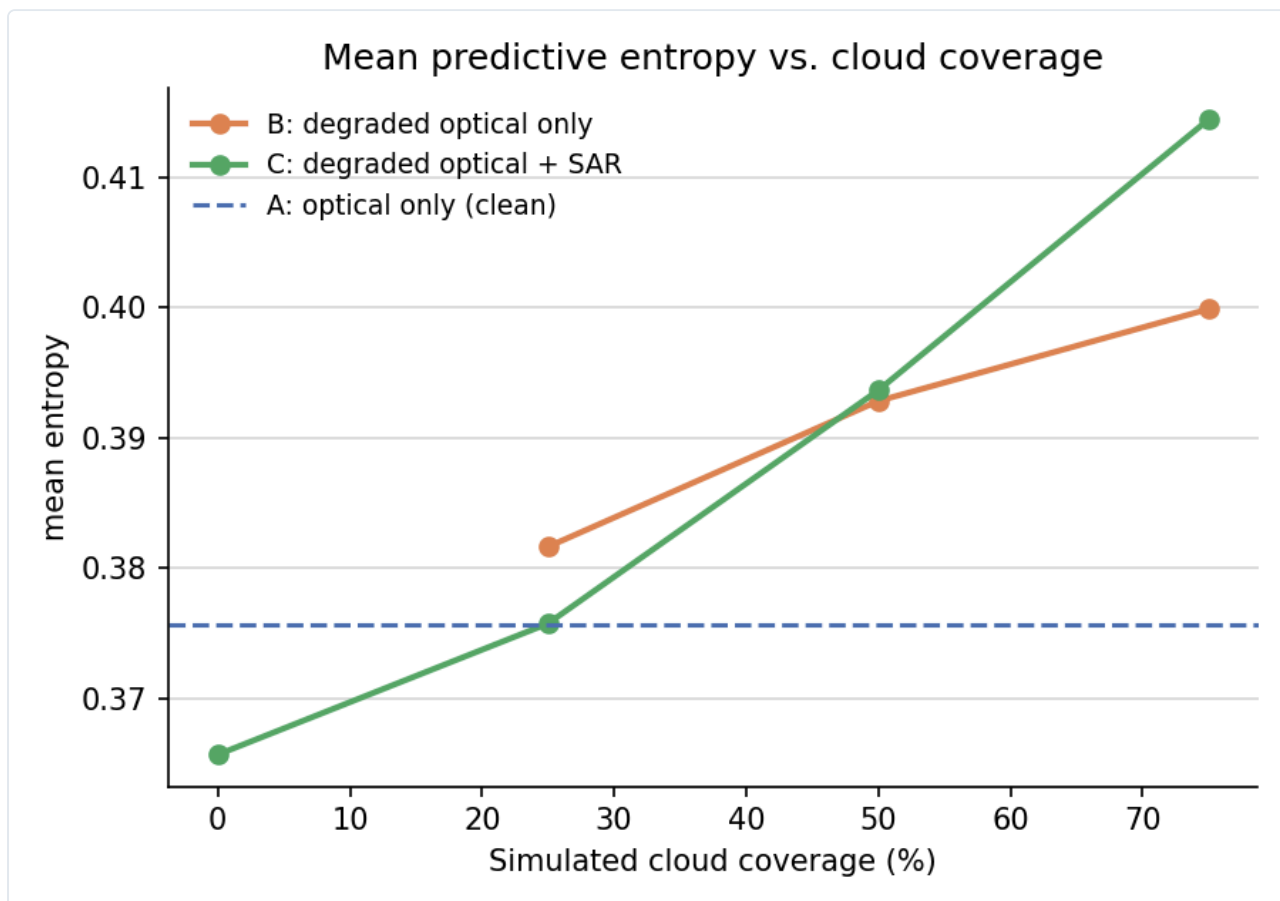


Figure 8. Mean predictive entropy vs. cloud coverage.

The optical-only model's mean entropy barely moves ($0.376 \rightarrow 0.400$) even as its macro-F1 collapses by half from clean to 75% coverage — it fails *confidently*, not appropriately uncertainly, the less useful failure mode for any system using predicted confidence to decide when to trust a result. The SAR-fusion model's entropy rises more with coverage ($0.366 \rightarrow 0.414$), a small but directionally correct signal that it "knows" the input is harder.

A single temperature is fit on **validation-set logits only**, then ECE is measured before/after on the test set, at 50% coverage:

Model	Temperature	ECE before	ECE after
B (optical only)	0.802	0.0335	0.0086
SAR-fusion	0.899	0.0236	0.0168

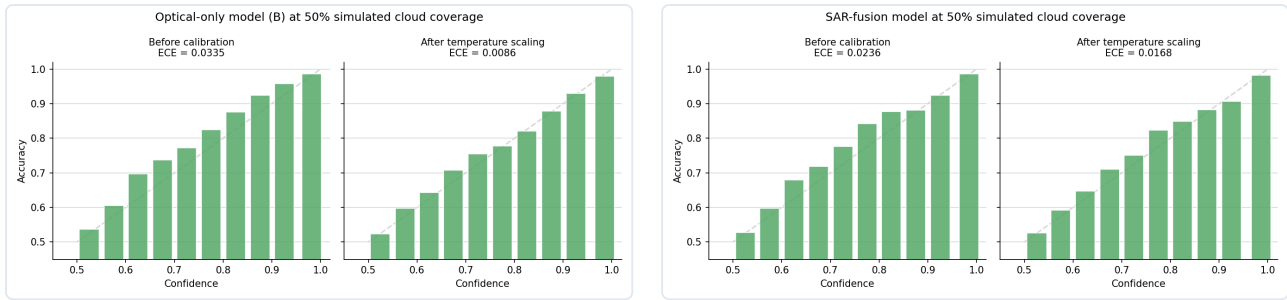


Figure 9. Reliability diagrams before/after temperature scaling: optical-only (left) and SAR-fusion (right), both at 50% coverage.

Both models are mildly *underconfident* at this operating point before correction (bars sit above the diagonal), and a temperature below 1 (which sharpens rather than softens probabilities) fixes this well for both. This refines rather than contradicts the entropy finding above: a model's probabilities can be reasonably well-calibrated in an absolute sense at one coverage level while still being insensitive to how input difficulty changes across coverage levels — both are true simultaneously here. The SAR-fusion model's calibration was already closer to ideal before correction, consistent with the idea that fusing an independent second modality moderates overconfidence somewhat.

11. Class-Confusion Analysis

Per-class F1 shows that a class is missed, not what the model says instead. Because this is a multi-label problem, a standard single-label confusion matrix doesn't directly apply; instead, for every test sample and every true class a model misses, every class it wrongly adds on that same sample is counted as a "confused-for" pair — computed at 50% coverage on the broad experiment's test set.

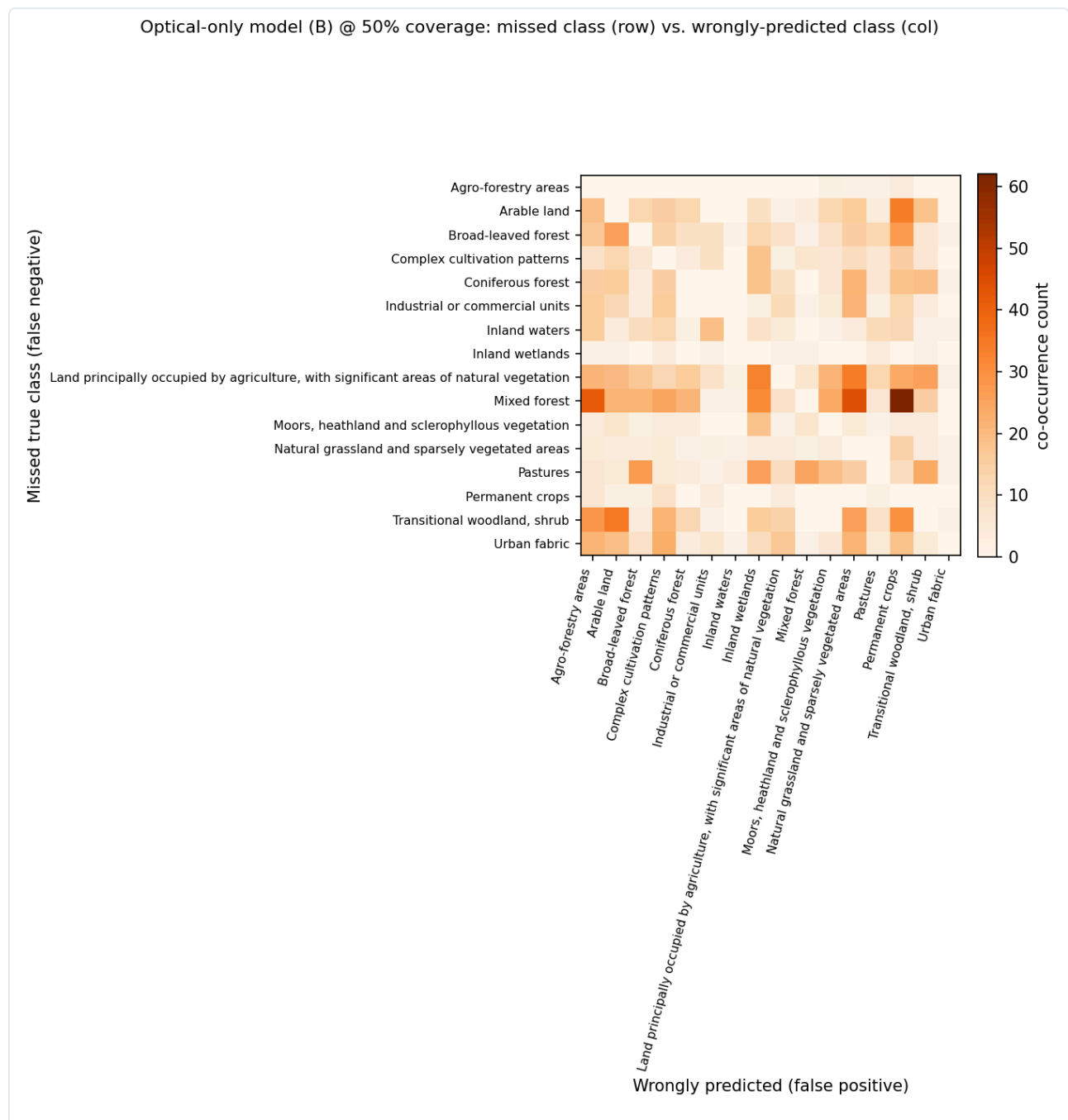


Figure 10a. Optical-only model confusion pattern at 50% coverage.

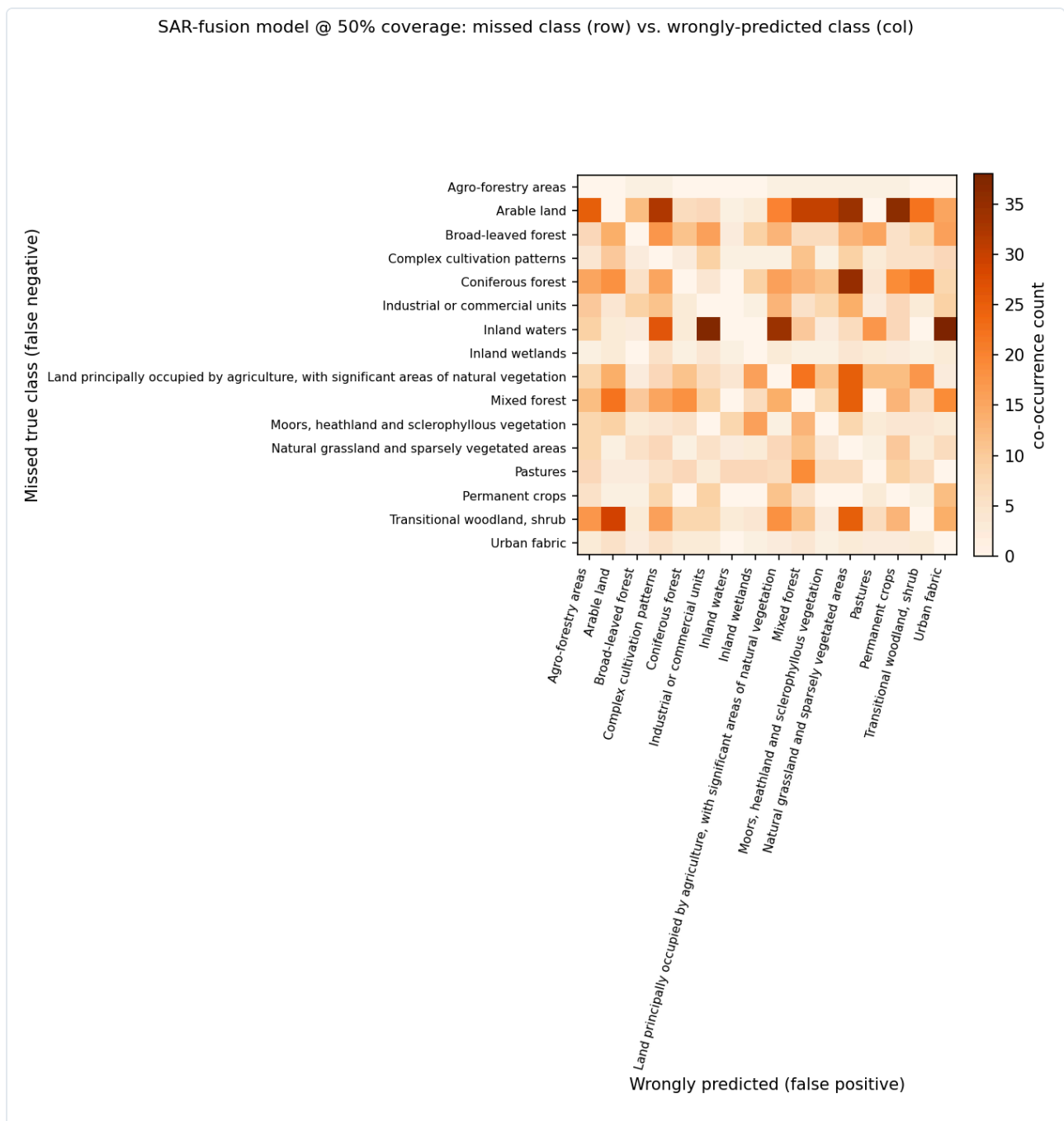


Figure 10b. SAR-fusion model confusion pattern at 50% coverage.

The optical-only model's dominant confusion is missing "Mixed forest" and instead predicting "Permanent crops," "Natural grassland," or "Agro-forestry areas" — semantically quite different land covers, consistent with the model falling back on whatever texture remains once enough of a forest patch is masked, rather than degrading toward a visually similar forest type.

The SAR-fusion model's confusion pattern is different in kind, not just smaller: its top confusion is missing "Inland waters" and predicting "Urban fabric" or "Industrial or commercial units" — a surprising water-vs-built-up mix-up with no obvious optical explanation. A plausible mechanism: calm inland water and certain urban/industrial surfaces can produce superficially similar low-texture SAR backscatter, and the fusion model may occasionally import this confusion rather than resolve it. This is a genuine, non-obvious failure mode that aggregate F1 numbers alone would never surface.

12. Honest Limitations and What's Next

In rough priority order, if given more time or compute budget:

1. **Multi-seed runs with confidence intervals.** Every number in this document is a single train/eval run. Section 7.2 showed a validation-set margin (0.0001) small enough that several finer-grained claims — exactly which fusion configuration is "best," the precise crossover coverage level — would benefit from 3–5 seeds per condition to know whether they survive scrutiny or are within run-to-run noise. This is the most legitimate point a careful reviewer would raise, and it is disclosed here rather than left for them to discover.
2. **A larger, more geographically diverse subset.** Both experiments use 5 tiles from 4 countries and an S1A-only pairing filter for bandwidth/time reasons. A broader subset would let "Beaches, dunes, sands" actually become learnable, and would let the depth-vs-breadth question in Section 9 be tested on a less geographically narrow sample.
3. **More realistic cloud simulation**, tested against BigEarthNet's own naturally-flagged cloud/shadow patches rather than synthetic rectangles.
4. **Calibration analysis across all four coverage levels** (this project only checked 50%), to see whether calibration degrades with coverage the way accuracy does.
5. **Map the depth-vs-breadth trade-off more finely** — e.g. 12–14 classes at an intermediate depth — rather than only the two endpoints tested here.

None of these gaps were hidden or discovered by omission — each is a deliberate scope decision made under the case's own explicit constraint not to over-invest in tuning or scale beyond what a 24–48 hour exercise calls for. Disclosing them plainly, rather than presenting a falsely complete picture, is itself part of the "scientifically valid evaluation" and "reasons about uncertainty" criteria this exercise asks for.

13. Project Artifacts and Reproduction

13.1 External resources used

- **Dataset:** BigEarthNet v2.0 / reBEN (Clasen et al., 2024), Zenodo record 10891137, CDLA-Permissive-1.0 license.
- **Libraries:** PyTorch & torchvision (models, training, LBFGS for temperature-scaling calibration), NumPy/pandas/pyarrow, scikit-learn, tiffle + Pillow, zstandard, matplotlib, requests.
- No pretrained model weights were used — every model was trained from scratch, since a 12/14-channel multispectral input does not align with standard 3-channel ImageNet-pretrained backbones without extra modification.

13.2 Reproducing this project end-to-end

```
pip install -r requirements.txt
python src/download_metadata.py # ~4.3 MB
python src/select_subset.py # broad-experiment selection (19 classes)
python src/extract_subset.py # streams ~1.8 GB from Zenodo
python src/preprocess.py # cached tensors + norm stats
python src/train_baseline_unweighted.py # unweighted-loss baseline (for the loss-abl
python src/run_broad_experiment.py # broad experiment: trains & evaluates
python src/confusion_analysis.py # class-confusion matrices (no retraining)

python src/select_subset_deep.py # deep-experiment selection (8 classes)
python src/extract_subset.py subset_patches_v3.csv # 2nd pass, same tiles, ~1.8 GB
python src/preprocess.py subset_patches_v3.csv _v3 # cache + norm stats
python src/run_deep_experiment.py # deep experiment + cross-experiment compa
python src/regenerate_final_figures.py # calibration/entropy figures (no retrainin
```

`train_baseline_unweighted.py` must run before `run_broad_experiment.py` for the class-balanced-loss ablation figure (Section 8.1) to be generated.

13.3 Repository layout

<code>src/</code>	all pipeline code, one script per pipeline stage
<code>data/</code>	metadata, subset selection CSVs for both experiments, raw extracted GeoTIFFs (excluded from version control, ~2.75 GB)
<code>outputs/cache/</code>	preprocessed tensors, shared across both experiments (excluded, ~7.9 GB)
<code>outputs/checkpoints/</code>	trained model weights (baseline + 3 fusion configurations, broad experiment; baseline + 1 fusion configuration, deep experiment)
<code>outputs/metrics/</code>	results tables, per-epoch training history, calibration and confusion summaries, cross-experiment comparison data
<code>outputs/figures/</code>	every figure referenced in this document and in README.md
<code>README.md</code>	top-level narrative and results (companion to this document)

REQUIREMENTS.md	full requirements-extraction pass from the case study PDF
DATA ACCESS.md	full data-access research trail (Section 3 of this document, in detail)

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(Zenodo 10891137)

